Surface Phenomena & Catalysis
Week 1: Bridge Module & Computational Perspectives

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Course Roadmap

Week	Chapter	Topic
1	**Ch 1–2**	**Bridge Module + Computational Perspectives**
2	Ch 3	Reactor Design & Catalyst Characterization
3	Ch 4	Transport Limitations
4	Ch 5	Selectivity Engineering
5	Ch 6	Zeolites & Carbon Nanotubes

Five weeks, six chapters—every lecture builds on the previous one.

Today's Agenda

Part I: Quick Recap *~15 min*

Bridge highlights + Ch 2 key models at a glance

Part II: Hands-On Notebooks *~25 min*

Guided tour of NB1 & NB2 — setup, highlights, exercises

Part III: Self-Study Guide *~5 min*

Notebook roadmap for Weeks 2–5 + resources

Lecture notes: Chapters 1–2 — Notebooks: NB1–NB2

Outline

1 Quick Recap: Bridge & Ch 2 Highlights

2 Hands-On Notebooks

3 Summary & Self-Study Guide

Five Steps of Heterogeneous Catalysis

1. **Diffusion to surface** — reactant molecules travel from bulk fluid to catalyst exterior
2. **Adsorption** — molecules bind to active sites ($A_{(g)} + * \rightleftharpoons A^*$)
3. **Surface reaction** — adsorbed species react to form products
4. **Desorption** — products leave the surface ($P^* \rightleftharpoons P_{(g)} + *$)
5. **Diffusion away** — products transport back to the bulk fluid

Design Implication

The *slowest* step determines the overall rate. We will learn how to identify it (Ch 3–4).

Essential Notation

Symbol	Meaning	Typical Units
X	Conversion	dimensionless
τ	Space time	s
r	Reaction rate	$\mathrm{mol\,m^{-3}\,s^{-1}}$
θ	Surface coverage	dimensionless
K	Adsorption equilibrium constant	Pa^{-1} (or bar^{-1})
k	Rate constant	depends on order
ϕ	Thiele modulus	dimensionless
η	Effectiveness factor	dimensionless

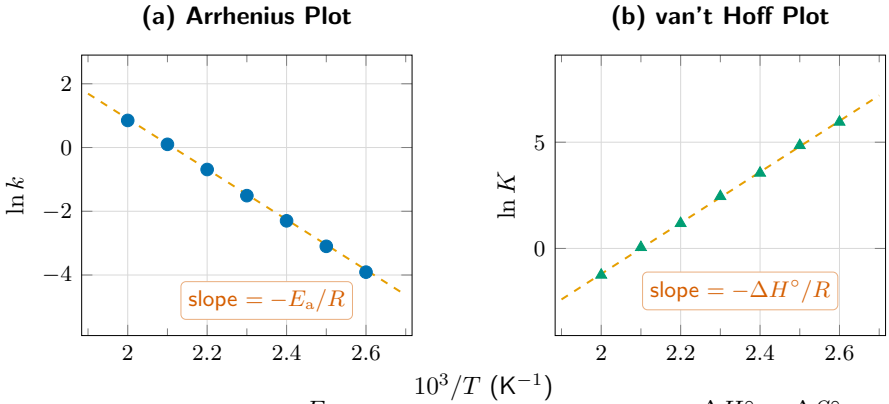
Full notation table: Appendix A of the lecture notes.

Key Equations at a Glance

- **Conversion:** $$X = \frac{C_{A0} - C_A}{C_{A0}}$$
- **Rate (per catalyst mass):** $$r = -\frac{1}{W}\frac{\mathrm{d}n_A}{\mathrm{d}t}$$ negative sign $\Rightarrow r > 0$
- **Site balance:** $$\theta_A + \theta_B + \cdots + \theta_* = 1$$
- **Langmuir isotherm:** $$\theta_A = \frac{K_A P_A}{1 + K_A P_A}$$

These four equations form the backbone of every derivation in this course.

k vs K — Rate Constant vs Equilibrium Constant



$$\ln k = \ln A - \frac{E_\mathrm{a}}{RT} \qquad \ln K = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R}$$

$k(T)$: **how fast?** $K(T)$: **how far?**

Three Mechanism Models at a Glance

Langmuir–Hinshelwood (LH) Both species adsorb, then react on surface

$$r = \frac{k\, K_A P_A\, K_B P_B}{(1 + K_A P_A + K_B P_B)^2}$$

Squared denominator $\Rightarrow$ rate shows a **maximum**, then declines (self-inhibition).

Eley–Rideal (ER) One adsorbed + one from gas phase

$$r = \frac{k_{\mathsf{ER}}\, K_A P_A\, P_B}{1 + K_A P_A}$$

First-order denominator $\Rightarrow$ monotonic saturation, no maximum.

Mars–van Krevelen (MVK) Redox cycle with lattice oxygen

$$\frac{1}{r} = \frac{1}{r_{\mathsf{red}}} + \frac{1}{r_{\mathsf{ox}}}$$

Two resistances in series $\Rightarrow$ slower half-cycle controls rate.

Mechanism Fingerprints

Diagnostic	**LH**	**ER**	**MVK**
r vs P_A shape	Maximum then decline	Monotonic saturation	Hyperbolic saturation
Denominator form	$(1+\sum K_i P_i)^2$	$(1+K_A P_A)$	$k_{\text{red}} P_A + k_{\text{ox}} P_{O_2}^n$
$E_{a,\text{app}}$ varies	Yes (with θ)	Yes (with θ)	Different E_{a} per step
Isotope test	$^{18}O_2$ no effect	—	$^{18}O_2$ in product

Key insight

Only LH shows a rate **maximum** with increasing P_A.

Competitive & Dissociative Isotherms

Competitive adsorption:

$$\theta_A = \frac{K_A P_A}{1 + \sum_j K_j P_j}$$

- All species share **same** sites
- Strong binder (poison) kills activity
- e.g. S on Pt, CO on fuel-cell Pt

Dissociative adsorption:

$$\theta_A = \frac{\sqrt{KP}}{1 + \sqrt{KP}}$$

- Molecule breaks on surface
- Needs **two adjacent** sites
- e.g. H_2 on Pt, N_2 on Fe

Details and derivations: Lecture Notes, Chapter 2.

Eyring Equation & Apparent E_a

Eyring (transition state theory):

$$k = \frac{k_\mathrm{B}T}{h}\, e^{\Delta S^\ddagger/R}\, e^{-\Delta H^\ddagger/RT}$$

- $\Delta H^\ddagger$: barrier height
- $\Delta S^\ddagger$: TS geometry (tight vs loose)
- Plot $\ln(k/T)$ vs $1/T$

Apparent activation energy:

Regime	$E_{a,\mathrm{app}}$
High θ	E_a
Low θ	$E_\mathrm{a} + \Delta H_\mathrm{ads}$

Since $\Delta H_\mathrm{ads} < 0$:
low θ gives **lower** apparent barrier.

NB2 Part III walks you through the Eyring analysis step by step.

Quick Concept Check

Concept Check

Three rapid-fire questions:

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3. $E_{a,\text{app}} < E_{\text{a}}$. Which coverage regime? **Low** θ ($E_{a,\text{app}} = E_{\text{a}} + \Delta H_{\text{ads}}$, with $\Delta H_{\text{ads}} < 0$)

Recap Summary

You know the models...

- **Adsorption:** Langmuir, competitive, dissociative isotherms
- **Mechanisms:** LH (rate max), ER (monotonic), MVK (redox cycle)
- **Temperature:** Arrhenius/Eyring, apparent vs intrinsic E_{a}
- **Diagnostics:** fingerprints, isotope labelling, operando

Now let's compute with them!

Next: hands-on tour of Notebooks NB1 and NB2.

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Getting Started: Jupyter & Google Colab

Option A: Local Jupyter

```
pip install notebook
jupyter notebook
```

- Full control, offline access
- Needs Python 3.9+ installed

Option B: Google Colab

1. Go to `colab.research.google.com`
2. Upload `.ipynb` file
3. Upload `data/` folder to session

- Zero install, runs in browser
- Free GPU if needed

Download notebooks and data files from `https://github.com/ExaPsi/2302337`.

NB1: Bridge Computational Tools

1. Read CSV → **2. Plot Data** → **3. Linear Fit** → **4. Residuals**

What you will learn:

- Load kinetic data from CSV
- Create publication-quality plots
- Perform linear regression with `linregress`
- Diagnose fit quality via residuals

Stats

47 cells, 3 exercises

Start here if you haven't used Python for kinetics before.

NB1 Highlight: Arrhenius Analysis Workflow

Four steps to extract $E_\mathbf{a}$ and A from data:

1. **Load CSV** with T and k columns
2. **Transform:** $x = 1/T$, $y = \ln k$
3. **Fit:** `linregress(x, y)`
4. **Extract:**
   $E_\mathrm{a} = -\text{slope} \times R$
   $A = \exp(\text{intercept})$

Report uncertainties from standard errors!

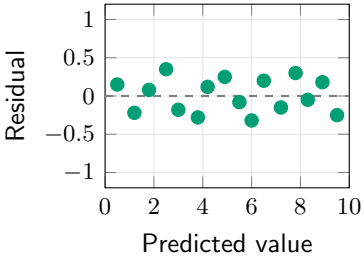
```
from scipy.stats import linregress

x = 1 / T_K
y = np.log(k)
res = linregress(x, y)

Ea = -res.slope * R
A  = np.exp(res.intercept)
```

NB1 Exercise 2 walks you through this step by step.

NB1 Highlight: Residual Analysis

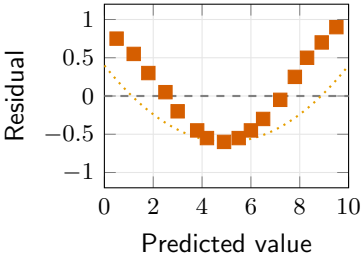
✓ **GOOD**

(a) Random Scatter



× **BAD**

(b) Systematic Pattern



Random scatter ⇒ good fit
Model captures the trend.

Systematic pattern ⇒ model deficiency
Try a different model or more parameters.

Always check residuals before trusting any fit.

NB2: Computational Perspectives on Surface Catalysis

Three parts, mirroring Chapter 2:

- Part I — Langmuir isotherms, competitive adsorption, TPD simulation
- Part II — LH/ER/MVK fingerprints, `curve_fit` for mechanism discrimination
- Part III — Arrhenius/Eyring parameter extraction, apparent E_a

Stats

70 cells, 6 exercises

This is the main computational companion for Ch 2.

Key Python tools

`scipy.optimize.curve_fit`

`numpy`, `matplotlib`

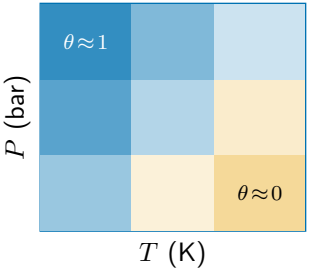
NB2 Highlight: Coverage Maps & Poisoning

Coverage map: $\theta(T, P)$ heatmap

- Visualise how T and P jointly control θ
- High K at low $T \Rightarrow$ saturated surface
- High $T \Rightarrow$ desorption wins, low θ

Competitive poisoning:

- Add inhibitor I with large K_I
- Watch θ_A collapse
- Exercise 6: design feed purity specs

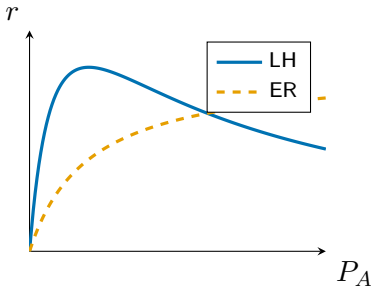


Schematic $\theta(T, P)$ heatmap

NB2 Highlight: Mechanism Discrimination

Workflow:

1. Generate synthetic rate vs P_A data
2. Fit **both** LH and ER models via `curve_fit`
3. Compare residuals: which model fits better?
4. The **squared denominator** makes the difference



Exercise 2

Fit LH and ER to the same dataset. Use residual plots to determine which mechanism generated the data.

Exercises to Complete This Week

NB1 (Bridge)

Ex	Topic	Priority
2	Arrhenius fitting	Essential
3	Residual analysis	Essential

NB2 (Comp. Perspectives)

Ex	Topic	Priority
1	Langmuir coverage	Essential
2	LH vs ER fitting	Essential
4	MVK analysis	Essential
6	Poisoning design	Bonus

Submission

Complete the **Essential** exercises and submit your notebooks via the course LMS before the start of Week 2. Notebooks and data files are available at `https://github.com/ExaPsi/2302337`.

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Key Equations Card

1. **Langmuir:** $\theta_A = K_A P_A \,/\, (1 + K_A P_A)$
2. **Competitive:** $\theta_A = K_A P_A \,/\, (1 + \sum K_j P_j)$
3. **Dissociative:** $\theta_A = \sqrt{KP} \,/\, (1 + \sqrt{KP})$
4. **LH:** $r = k\, K_A P_A\, K_B P_B \,/\, (1 + K_A P_A + K_B P_B)^2$
5. **ER:** $r = k_{\mathsf{ER}}\, K_A P_A\, P_B \,/\, (1 + K_A P_A)$
6. **MVK:** $1/r = 1/r_{\mathsf{red}} + 1/r_{\mathsf{ox}}$
7. **Eyring:** $k = (k_{\mathrm{B}} T/h)\, e^{\Delta S^{\ddagger}/R}\, e^{-\Delta H^{\ddagger}/RT}$
8. **Apparent $E_{\mathbf{a}}$ (low θ):** $E_{a,\mathsf{app}} = E_{\mathrm{a}} + \Delta H_{\mathsf{ads}}$

Master these eight—they form the foundation for all that follows.

Self-Study Notebook Roadmap

Week	Chapter	Notebook	Key Skills
1	**Ch 1–2**	**NB1, NB2**	**CSV, plotting, Langmuir, LH/ER/MVK fitting**
2	Ch 3	NB3	CSTR/PFR solvers, BET analysis
3	Ch 4	NB4	Thiele modulus, effectiveness factor
4	Ch 5	NB5	Yield optimisation simulations
5	Ch 6	NB6	Zeolite/CNT transport modelling

How to use the notebooks

- Run **all cells** first to see the expected output
- Read the narrative cells—they explain the "why"
- Then attempt the exercises; check your residuals!

Resources & Links

GitHub Repository

`https://github.com/ExaPsi/2302337`

Lecture Notes Full PDF
6 chapters + appendices

Notebooks NB1–NB6 + NB-S
`.ipynb` files + `data/` folder

`utils.py` Shared functions
Langmuir, Arrhenius, LH rate, reactor solvers

Getting Help

- Course LMS discussion board
- Office hours: TBA
- Google Colab for zero-install Python

Tip

Import shared functions with:

```
from utils import *
```

Looking Ahead — Week 2

Next: Reactor Design & Catalyst Characterisation (Chapter 3)

- Batch, CSTR, PFR design equations
- Turnover frequency (TOF) and turnover number (TON)
- BET surface area, TPD, XRD, TEM, FTIR

Preparation

- Read Chapter 3, Sections 3.1–3.4
- Complete NB1 & NB2 exercises
- Review site balance and Langmuir

Key question for Week 2

Given a rate law, how do we **size a reactor** to achieve a target conversion?

See you next week!